Team Task Board

Flutter Integration Spec - Dovix AI Employee App

Written for: the Flutter developer building the Team Task Board screens in the mobile app.

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1. Overview

This is the same Team Task Board already live in the web admin panel (Teams → pick a team → board) - an Azure-DevOps-style Kanban board scoped to a team, with sprints, task types (Task/Bug), severity, project tagging, comments, and image/video attachments. The backend API for it is fully built and tested; this document covers the screens to build and the exact endpoints that feed them.

Screens to build:

- **My Teams** - list of teams you're on, tap one to open its board.
- **Team Board** - sprint tabs, member filter chips, status columns, insight counts, a “+” to quick-add.
- **Task Detail** - editable fields (title, description, status, type, severity, assignee, sprint, project), an attachment gallery, and a comment thread with its own attach button.

Everything on this board already pushes a notification through the Firebase integration delivered earlier (task assigned/reassigned) - nothing extra is needed for that here.

2. Adapting the Web Board's UI to Mobile

The web board is a desktop drag-and-drop grid (a collapsible row per member, sprint tabs, status columns). That doesn't translate directly to a phone screen - here's the mobile-native equivalent for each piece, same underlying capability, touch-first interaction:

Web board	Mobile equivalent
Sprint tabs	Top segmented tab bar / horizontal chip row (Backlog, Sprint 1, Sprint 2, ...)
Collapsible member rows (swimlanes)	Horizontal avatar filter chips above the columns (“All”, “Unassigned”, each member) - filters which cards show
4 status columns	Same 4 columns (To Do / In Progress / Blocked / Done), horizontally swipeable, same colors
Drag a card between columns	Tap the card → Task Detail screen → status picker (or a quick long-press action sheet if you prefer)
Side panel (edit everything)	The Task Detail screen - same fields, saved through one PATCH call
“+ New work item” under To Do	A “+” FAB on the board opening a small quick-add sheet

This is a suggested layout, not a strict requirement - keep whatever a real board screen in this app already does for navigation/tabs conventions; the API doesn't care how you present it, only what you send.

3. Data Model Reference

Every enum value the API uses, exactly as sent/expected (all uppercase over the wire):

- **status**: PENDING (“To Do”), IN_PROGRESS, BLOCKED, COMPLETED (“Done”)
- **type**: TASK, BUG
- **severity**: LOW, MEDIUM, HIGH, CRITICAL - only meaningful when type is BUG
- **priority**: LOW, MEDIUM, HIGH, CRITICAL - separate from severity, shown as the card's left-border color

Sprint / Backlog: a team's branch can have a sprint cadence configured (e.g. 2-week sprints). A task's `sprintId` is null/omitted when it's in the **Backlog** (not yet scheduled into a sprint) - the board's own “Backlog” tab is just `?sprint=backlog`. If a branch has no sprint configuration yet, `sprints` in the board response is simply empty and everything lives in Backlog.

Team membership: any member of a team (not just its manager) can view the board, drag/change status, reassign to any other member, and comment - there's no separate “read-only member” role.

4. API Reference

All endpoints require `Authorization: Bearer <accessToken>` (same JWT as the rest of the app) and return the standard envelope `{success, data, message, error}`. Full request/response examples for each are in api/docs (Teams section) - this is the quick map plus the payload shapes you'll build your models from.

Method & Path	Purpose
`GET /api/v1/teams`	Teams you're a member of
`GET /api/v1/teams/{id}/board`	The board screen's one big fetch (sprints, members, projects, insights, tasks)
`POST /api/v1/teams/{id}/board/tasks`	Quick-add a task/bug
`GET /api/v1/teams/tasks/{id}`	One task's full detail + activity feed
`PATCH /api/v1/teams/tasks/{id}`	Update any field(s) - status move, reassign, edit, etc.
`POST /api/v1/teams/tasks/{id}/comments`	Post a comment, optional image/video
`POST /api/v1/teams/tasks/{id}/attachments`	Attach media to the task itself (not a comment)

4.1 GET /teams


```
{
  "teams": [
    {"id": 3, "name": "Backend", "description": "API + infra", "manager": "Akshay",
     "memberCount": 4, "taskCount": 12, "canManage": false}
  ]
}
```


Team ids here are plain integers (not prefixed like task/project ids) - use them directly in the board URL.

4.2 GET /teams/{id}/board?sprint=&project=

`sprint`: a sprint id, or `"backlog"` - omit it to default to whichever sprint covers today (or Backlog if none configured). `project`: optional project id filter, e.g. `"prj_00003"`.


```
{
  "team": {"id": 3, "name": "Backend", "description": "API + infra", "canManage": false},
  "members": [{"id": 14, "name": "Kamal"}, {"id": 13, "name": "Akshay"}],
  "sprints": [
    {"id": 4, "name": "Sprint 1", "startDate": "2026-09-13", "endDate": "2026-09-26",
     "isCurrent": true, "isSelected": true},
    {"id": 5, "name": "Sprint 2", "startDate": "2026-09-27", "endDate": "2026-10-10",
     "isCurrent": false, "isSelected": false}
  ],
  "selectedSprint": "4",
  "projects": [{"id": "prj_00004", "code": "PRJ00004", "name": "Finance Automation"}],
  "insights": {"total": 5, "pending": 2, "inProgress": 2, "blocked": 0, "completed": 1},
  "tasks": [
    {"id": "tsk_00016", "title": "Fix login bug", "description": "Repro on Safari",
     "status": "PENDING", "type": "BUG", "severity": "HIGH", "priority": "MEDIUM",
```

```
     "assignedTo": {"id": 14, "name": "Kamal"}, "project": null, "projectColor": null,
     "sprintId": 4, "dueDate": null, "commentCount": 1, "createdAt":
"2026-09-16T08:37:41Z"}
  ]
}
```

Group `tasks` into your own To Do/In Progress/Blocked/Done columns (by `status`) and member filter (by `assignedTo.id`, null = Unassigned) client-side - the same data the web board groups server-side, just handed over flat.

4.3 POST /teams/{id}/board/tasks

multipart or form-encoded. Left unassigned unless you set assignedTo.

```
title=Fix login bug
type=BUG
severity=HIGH
assignedTo=14 // optional - a member id from this team
project=prj_00003 // optional
sprint=4 // optional - omit for Backlog
```

Response: `{"id": "tsk_00021"}` - fetch full detail with 4.4 if needed.

4.4 GET /teams/tasks/{id}

Same task shape as the board list, plus:

```
{
  ... (all fields from the board list) ...
  "attachments": [{"url": "/media/task_attachments/16/repro.png", "kind": "IMAGE"}],
  "activity": [
    {"author": "Akshay", "comment": null, "previousStatus": null, "newStatus": "PENDING",
     "createdAt": "2026-09-16T08:37:41Z", "attachments": []},
    {"author": "Kamal", "comment": "Repro'd on Safari 17", "previousStatus": "PENDING",
     "newStatus": "PENDING", "createdAt": "2026-09-16T09:10:02Z",
     "attachments": [{"url": "/media/task_attachments/16/video.mp4", "kind": "VIDEO"}]}
  ]
}
```

`activity` is the full history, oldest first - render it as a chat/timeline feed under the task's editable fields. A pure status/field change has `comment: null`; a real comment has text and/or an attachment.

4.5 PATCH /teams/tasks/{id}

The one endpoint behind every edit on the Task Detail screen. Send only the field(s) that changed - JSON body, not multipart.

```
// Drag-equivalent: just move the status
{"status": "IN_PROGRESS"}

// Reassign to someone else on the team
{"assignedTo": "14"}

// Explicitly unassign / move to Backlog / remove from a project - empty string, not
omitted
{"assignedTo": "", "sprint": "", "project": ""}
```

```
// Several fields from the detail screen at once
{"title": "Fix login bug on Safari", "description": "Users cannot log in.",
 "severity": "CRITICAL", "sprint": "5"}
```

Response is the updated task (same shape as 4.2's list items). Every change - status, reassignment, sprint/project move, edits - is recorded in the activity feed automatically, with who and when; you don't need to post a separate comment to log it.

4.6 POST /teams/tasks/{id}/comments

multipart/form-data (not JSON) so an image or short video can ride along - e.g. a screen recording showing a bug happening. At least one of `comment` / `attachment` is required.

```
comment=Seeing this on Safari too
attachment=<file> // optional - image or short video

{
  "author": "Kamal", "comment": "Seeing this on Safari too",
  "createdAt": "2026-09-16T09:10:02Z",
  "attachment": {"url": "/media/task_attachments/16/video.mp4", "kind": "VIDEO"}
}
```

4.7 POST /teams/tasks/{id}/attachments

multipart/form-data, `attachment` required. Attaches directly to the task's own description gallery, not tied to a specific comment - use this for the “add repro evidence to the bug itself” action versus commenting.

```
{"url": "/media/task_attachments/16/screenshot.png", "kind": "IMAGE"}
```

File limits (both endpoints): images (JPG/PNG/GIF/WEBP) up to 8MB, short videos (MP4/WEBM/MOV) up to 30MB.

5. Testing Checklist

1. Log in, GET /teams - confirm every team you're on appears.
2. Open a team's board - confirm sprint tabs, member chips, columns, and insight counts all populate correctly.
3. Quick-add a Bug with a severity and an assignee; confirm it appears in the right column/member filter.
4. Move a card's status via the detail screen; confirm the activity feed logs it.
5. Reassign a task to a teammate; confirm they get a push notification (Firebase integration).
6. Post a comment with an attached image and one with an attached short video; confirm both render correctly in the activity feed.
7. Attach media directly to a task's description (not via a comment) and confirm it shows in the task's attachment gallery, separate from the comment thread.
8. Switch sprint tabs and the project filter; confirm the task list narrows correctly for each.

Questions on the backend side (payload shape, adding a new field, timing) - ask the backend developer; this document covers only what's needed in the Flutter app itself.